Lab 2: The Value of Prediction — Answer Key

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ECON 470 — Microeconomics of Artificial Intelligence

Spring 2026

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1 Problem 1: Medical Diagnosis and the Value of Prediction

Parameters: $p = 0.7$, $R = 100$, $r = 20$, $e = 0.85$.

1.1 Part A: Is the signal informative?

Q1. Joint probability table $\Pr(\theta = i, s = j)$:

Using Table 3.2 in Gans with $e_1 = e_2 = e = 0.85$:

	$s = 1$	$s = 2$
$\theta = 1$	$ep = 0.85 \times 0.7 = 0.595$	$(1-e)p = 0.15 \times 0.7 = 0.105$
$\theta = 2$	$(1-e)(1-p) = 0.15 \times 0.3 = 0.045$	$e(1-p) = 0.85 \times 0.3 = 0.255$

Check: $0.595 + 0.105 + 0.045 + 0.255 = 1.000$. ✓

Q2. Posterior given positive signal:

$$\Pr(\theta = 1 \mid s = 1) = \frac{\Pr(\theta = 1, s = 1)}{\Pr(s = 1)} = \frac{0.595}{0.595 + 0.045} = \frac{0.595}{0.640} \approx 0.9297$$

Since $0.9297 > 0.7 = p$, receiving $s = 1$ makes the physician **more** confident the infection is present.

Q3. Posterior given negative signal:

$$\Pr(\theta = 1 \mid s = 2) = \frac{\Pr(\theta = 1, s = 2)}{\Pr(s = 2)} = \frac{0.105}{0.105 + 0.255} = \frac{0.105}{0.360} \approx 0.2917$$

Since $0.2917 < 0.7 = p$, receiving $s = 2$ makes the physician **less** confident the infection is present (equivalently, more confident it is not present).

Q4. The signal is "informative" because it moves the physician's beliefs in the correct direction: $s = 1$ pushes the posterior above the prior (toward $\theta = 1$), and $s = 2$ pushes the posterior below the prior (toward $\theta = 2$). This is guaranteed whenever $e_1 + e_2 > 1$.

1.2 Part B: Expected payoff without a signal

Q5. Since $p = 0.7 > 0.5$, state 1 is more likely, so the best rule is $a = 1$ (treat):

$$\text{Payoff}_{a=1} = pR + (1-p)r = 0.7 \times 100 + 0.3 \times 20 = 70 + 6 = 76$$

The alternative rule $a = 2$ (always not treat):

$$\text{Payoff}_{a=2} = pr + (1-p)R = 0.7 \times 20 + 0.3 \times 100 = 14 + 30 = 44$$

1.3 Part C: Strategies with a signal

Q6. The four strategies are maps from $\{s = 1, s = 2\} \to \{a = 1, a = 2\}$:

Strategy	$s = 1$	$s = 2$	Description
S1: Always $a = 1$	$a = 1$	$a = 1$	Ignore the signal; always treat
S2: Always $a = 2$	$a = 2$	$a = 2$	Ignore the signal; never treat
S3: Follow signal	$a = 1$	$a = 2$	Treat if positive, don't treat if negative
S4: Oppose signal	$a = 2$	$a = 1$	Don't treat if positive, treat if negative

Q7. For each strategy, the payoff in each (θ, s) cell:

Strategy S1 (always $a = 1$):

	$s = 1$	$s = 2$
$\theta = 1$	$R = 100$	$R = 100$
$\theta = 2$	$r = 20$	$r = 20$

$E[U_{S1}] = 0.595(100) + 0.105(100) + 0.045(20) + 0.255(20) = 59.5 + 10.5 + 0.9 + 5.1 = 76$

Strategy S2 (always $a = 2$):

	$s = 1$	$s = 2$
$\theta = 1$	$r = 20$	$r = 20$
$\theta = 2$	$R = 100$	$R = 100$

$E[U_{S2}] = 0.595(20) + 0.105(20) + 0.045(100) + 0.255(100) = 11.9 + 2.1 + 4.5 + 25.5 = 44$

Strategy S3 (follow signal):

	$s = 1$	$s = 2$
$\theta = 1$	$R = 100$ (match)	$r = 20$ (mismatch)
$\theta = 2$	$r = 20$ (mismatch)	$R = 100$ (match)

$E[U_{S3}] = 0.595(100) + 0.105(20) + 0.045(20) + 0.255(100) = 59.5 + 2.1 + 0.9 + 25.5 = 88$

Strategy S4 (oppose signal):

	$s = 1$	$s = 2$
$\theta = 1$	$r = 20$ (mismatch)	$R = 100$ (match)
$\theta = 2$	$R = 100$ (match)	$r = 20$ (mismatch)

$E[U_{S4}] = 0.595(20) + 0.105(100) + 0.045(100) + 0.255(20) = 11.9 + 10.5 + 4.5 + 5.1 = 32$

Summary:

Strategy	Expected Payoff
S1: Always treat	76
S2: Never treat	44
S3: Follow signal	**88**
S4: Oppose signal	32

Q8. Strategy S3 (follow the signal) yields the highest expected payoff of 88. This is consistent with $e = 0.85 \geq p = 0.7$: the signal is accurate enough to beat the prior, so following it is optimal. Note that S1 and S2 are the two "rules" that ignore the signal, and both yield the same payoffs as Part B.

1.4 Part D: The value of the signal

Q9.

$$V(e) = 88 - 76 = 12$$

Check: $(e - p)(R - r) = (0.85 - 0.7)(100 - 20) = 0.15 \times 80 = 12$. ✓

2 Problem 2: The Demand for AI Diagnostic Services

Parameters: $p = 0.7$, $e = 0.85$, $R_i - r_i \sim \text{Uniform}[10, 200]$, 1,000 physicians.

2.1 Part A: Willingness to pay

Q10. The value of the signal to physician i is:

$$\text{WTP}_i = (e - p)(R_i - r_i) = 0.15(R_i - r_i)$$

Q11.

- Lowest stakes ($R_i - r_i = 10$): WTP $= 0.15 \times 10 = 1.50$
- Highest stakes ($R_i - r_i = 200$): WTP $= 0.15 \times 200 = 30$
- Problem 1 physician ($R_i - r_i = 80$): WTP $= 0.15 \times 80 = 12$

2.2 Part B: Demand curve

Q12. A physician adopts if $\text{WTP}_i \geq P$:

$$0.15(R_i - r_i) \geq P \implies R_i - r_i \geq \frac{P}{0.15} = \frac{20P}{3}$$

Q13. The fraction of physicians with $R_i - r_i \geq \frac{P}{0.15}$ under a Uniform$[10, 200]$ distribution:

$$D(P) = 1000 \times \frac{200 - P/0.15}{200 - 10} = 1000 \times \frac{200 - 20P/3}{190}$$

This is valid when the cutoff $P/0.15$ falls in $[10, 200]$, i.e., $P \in [1.50, 30]$.

- For $P < 1.50$: $D(P) = 1000$ (all physicians adopt)
- For $P > 30$: $D(P) = 0$ (no one adopts)
- For $P \in [1.50, 30]$: $D(P)$ is linear and decreasing

Q14. See figure below. The demand curve is a downward-sloping line from $(P, D) = (1.50, 1000)$ to $(30, 0)$.

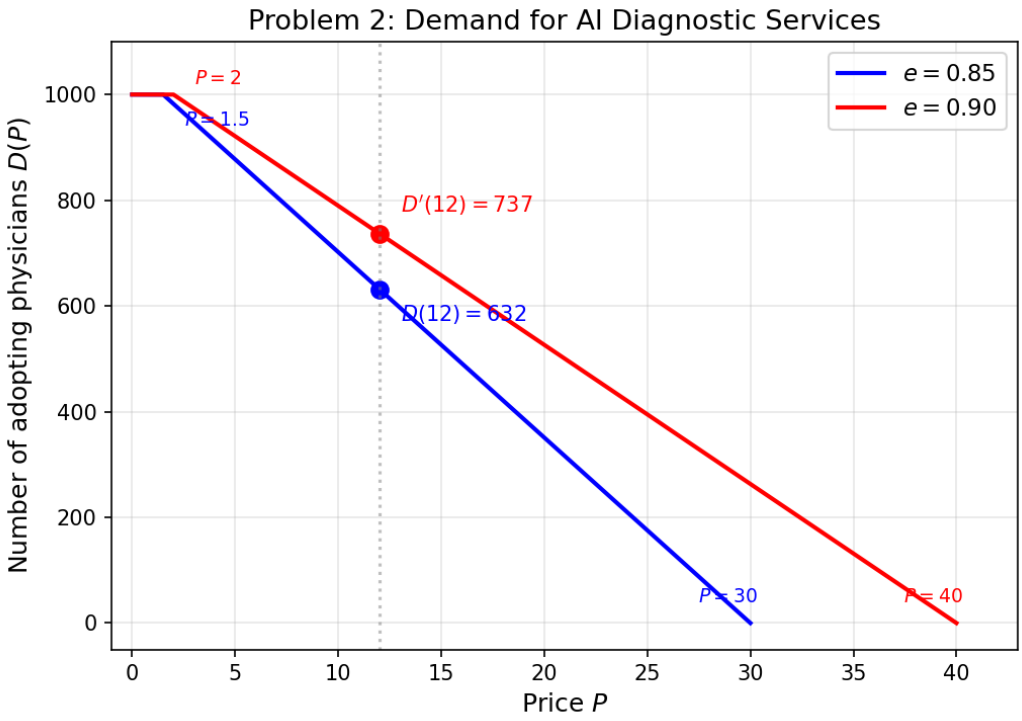


Figure 1: Demand curves for AI diagnostic services.

2.3 Part C: Better AI

Q15. With $e = 0.90$: $\text{WTP}_i' = (0.90 - 0.7)(R_i - r_i) = 0.20(R_i - r_i)$.

Cutoff: $R_i - r_i \geq P/0.20 = 5P$.

$$D'(P) = 1000 \times \frac{200 - 5P}{190}$$

Valid for $P \in [2, 40]$. Plotted in red in the figure above.

Q16. At $P = 12$:

- Old AI ($e = 0.85$): cutoff $= 12/0.15 = 80$, so $D(12) = 1000 \times \frac{200-80}{190} = 1000 \times \frac{120}{190} \approx 632$
- New AI ($e = 0.90$): cutoff $= 12/0.20 = 60$, so $D'(12) = 1000 \times \frac{200-60}{190} = 1000 \times \frac{140}{190} \approx 737$

About 105 additional physicians adopt the better AI at the same price.

Q17. Better prediction increases the WTP of every physician because $V(e) = (e - p)(R - r)$ is increasing in e. A higher e means the signal resolves more uncertainty, so each physician's willingness to pay rises. At any given price P, the cutoff $R_i - r_i \geq P/(e - p)$ is lower with higher e, so more physicians clear the bar.

3 Problem 3: Biased Signals and the Spam Filter

Parameters: $p = 0.95$, $R = 1$, $r = 0.5$, $\bar{\lambda} = 0.70$, $e_1 + e_2 = 1.40$.

3.1 Part A: Expected payoff as a function of e_1 and e_2

Q18. The agent follows the signal: $a = 1$ when $s = 1$, $a = 2$ when $s = 2$. The payoff in each cell:

	$s = 1 \to a = 1$	$s = 2 \to a = 2$
$\theta = 1$	$R = 1$ (match)	$r = 0.5$ (mismatch)
$\theta = 2$	$r = 0.5$ (mismatch)	$R = 1$ (match)

Using joint probabilities from Table 3.3:

$$\begin{aligned} E[U] &= e_1 p \cdot R + (1 - e_1)p \cdot r + (1 - e_2)(1 - p) \cdot r + e_2(1 - p) \cdot R \\ &= e_1(0.95)(1) + (1 - e_1)(0.95)(0.5) + (1 - e_2)(0.05)(0.5) + e_2(0.05)(1) \\ &= 0.95e_1 + 0.475 - 0.475e_1 + 0.025 - 0.025e_2 + 0.05e_2 \\ &= 0.475e_1 + 0.025e_2 + 0.5 \end{aligned}$$

The expected payoff is **linear** in e_1 and e_2, with the coefficient on e_1 (0.475) being **19 times larger** than the coefficient on e_2 (0.025). This reflects the fact that $p = 0.95$: most messages are legitimate, so accuracy in state 1 matters far more than accuracy in state 2.

3.2 Part B: The accuracy frontier

Q19–Q21. The constraint $e_1 + e_2 = 1.40$ with $e_i \in [0, 1]$ is a line segment from $(e_2, e_1) = (0.40, 1.00)$ to $(1.00, 0.40)$.

The indifference curves of $U = 0.475e_1 + 0.025e_2 + 0.5$ are straight lines with slope $-0.025/0.475 = -1/19$ (nearly flat). Payoff increases in the direction of higher e_1 (upward).

The optimum is at the **upper-left corner** of the feasible segment: $e_1^* = 1.00$, $e_2^* = 0.40$.

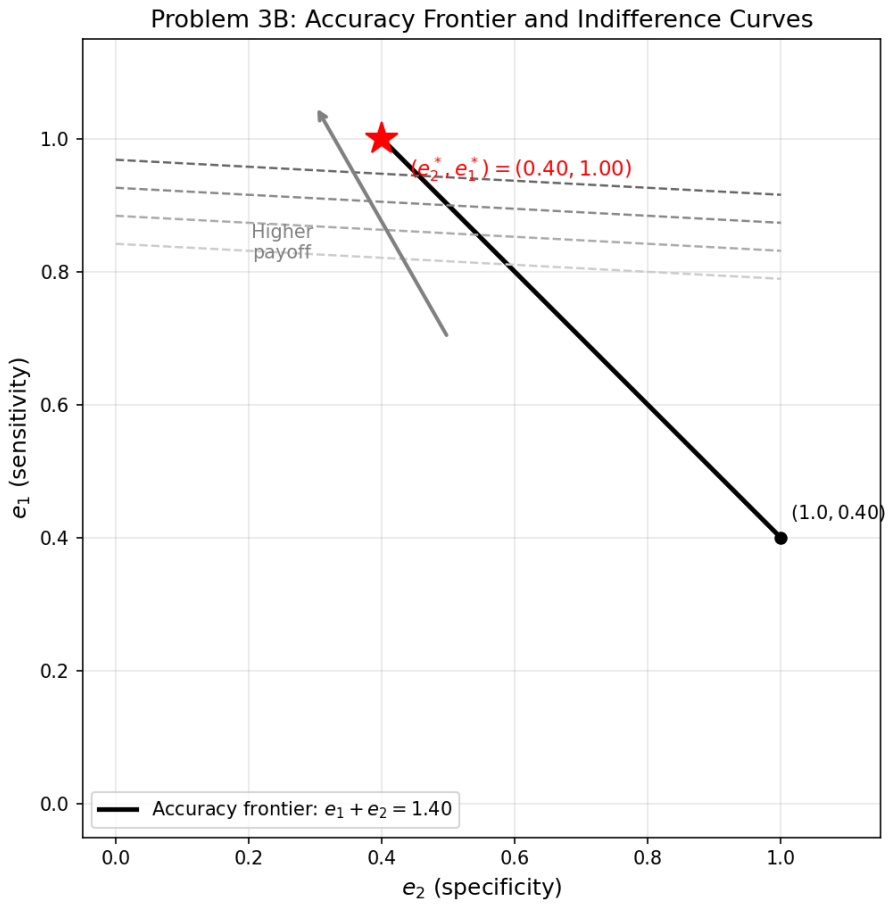


Figure 2: Accuracy frontier and indifference curves.

3.3 Part C: Interpretation

Q22. The agent prioritizes **sensitivity over specificity**. With $e_1 = 1$, the filter never flags a legitimate email as spam (no false negatives). The cost is low specificity ($e_2 = 0.40$): 60% of spam messages slip through. But since only 5% of messages are spam ($1 - p = 0.05$), this amounts to $0.60 \times 0.05 = 3\%$ of all messages being misclassified spam that gets through — a tolerable cost compared to the alternative of losing legitimate emails.

Q23. Without a signal, the default is $a = 1$ (mark everything as not-spam).

- **False positive** (spam gets through: $\theta = 2$, signal says $s = 1$, agent does $a = 1$): the agent was going to choose $a = 1$ anyway. Incremental cost relative to default: **zero**.
- **False negative** (legitimate email flagged as spam: $\theta = 1$, signal says $s = 2$, agent switches to $a = 2$): the agent would have gotten it right with the default, but the signal caused the wrong action. Incremental cost relative to default: $R - r = 0.5$.

Since only false negatives impose an incremental cost, the agent eliminates them entirely by setting $e_1 = 1$.

3.4 Part D: The ROC curve

Q24. With $\text{FPR} = 1 - e_2$ and $\text{TPR} = e_1$, the constraint $e_1 + e_2 = 1.40$ becomes:

$$\text{TPR} = 1.40 - (1 - \text{FPR}) = 0.40 + \text{FPR}$$

This is a line segment from $(\text{FPR}, \text{TPR}) = (0, 0.40)$ to $(0.60, 1.00)$.

The indifference curves become $U = 0.475 \cdot \text{TPR} - 0.025 \cdot \text{FPR} + 0.525$, which are nearly flat lines (slope $= 0.025/0.475 = 1/19$). Payoff increases upward (higher TPR).

The optimum is at $(\text{FPR}^*, \text{TPR}^*) = (0.60, 1.00)$ — the upper-right endpoint.

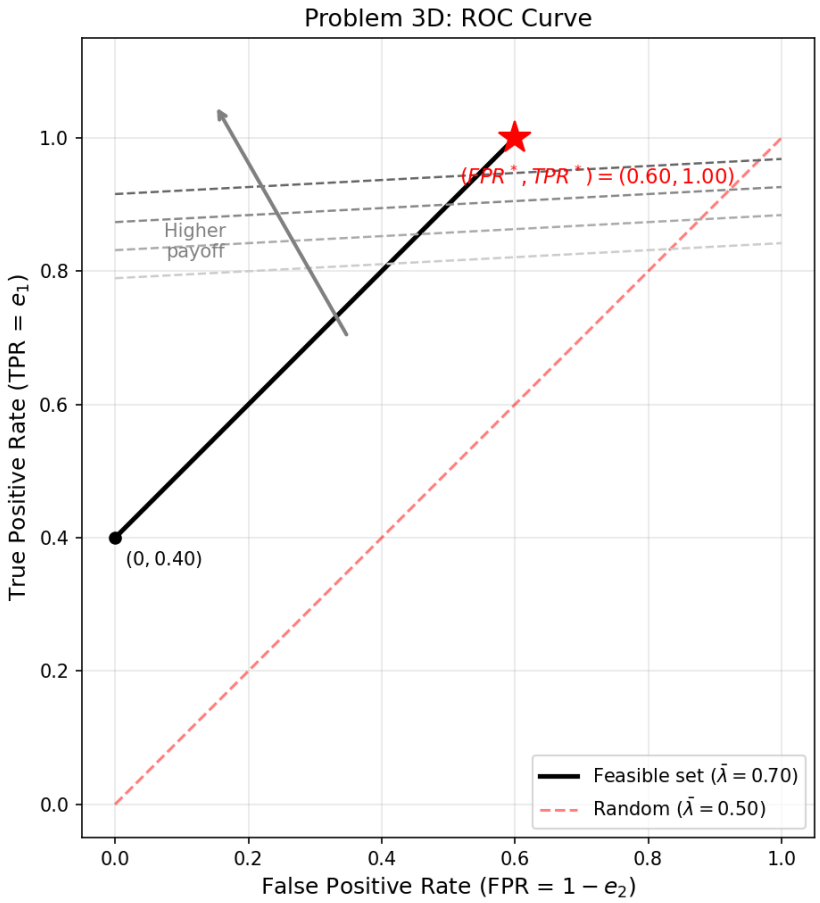


Figure 3: ROC curve with optimal point.

Q25. The accuracy frontier in the ROC diagram is the line segment of feasible (FPR, TPR) pairs. It plays the same role as the budget constraint in consumer theory: the agent picks the point on the frontier tangent to the highest indifference curve.

Improving the prediction technology (increasing $\bar{\lambda}$) shifts the frontier **toward the upper-left corner** $(0, 1)$: the lower endpoint $(0, 2\bar{\lambda} - 1)$ slides up the vertical axis and the upper endpoint

$(2-2\bar{\lambda}, 1)$ slides left along the top of the unit square. The agent gets access to higher TPR at every level of FPR, expanding the set of feasible error trade-offs.

4 Problem 4: Reading the ROC Curve

Q26. The red line (random prediction). If $e_1 + e_2 = 1$, then $e_2 = 1 - e_1$, so:

$$\text{TPR} = e_1 \quad \text{and} \quad \text{FPR} = 1 - e_2 = 1 - (1 - e_1) = e_1$$

Therefore TPR = FPR for every point on this line.

A signal where TPR = FPR is useless because it "detects" the positive state at exactly the same rate as it falsely "detects" it when it is absent. The signal fires with the same frequency regardless of the true state, so it carries no information — it cannot distinguish $\theta = 1$ from $\theta = 2$. Receiving the signal does not update the agent's beliefs.

Q27. The green line (perfect prediction). At the upper-left corner $(0, 1)$: FPR = 0 means $e_2 = 1$ (the signal never says $s = 1$ when $\theta = 2$), and TPR = 1 means $e_1 = 1$ (the signal always says $s = 1$ when $\theta = 1$). Together, $e_1 = e_2 = 1$: the signal is perfectly accurate in both states. This is the maximum $\bar{\lambda} = 1$.

When $\bar{\lambda} = 1$, the constraint $e_1 + e_2 = 2$ with $e_i \in [0, 1]$ forces $e_1 = e_2 = 1$, so the feasible set collapses to the single point $(0, 1)$. The green L-shape in the figure is the **limit of the accuracy frontier** as $\bar{\lambda} \to 1$: the lower endpoint $(0, 2\bar{\lambda} - 1)$ rises to $(0, 1)$ and the upper endpoint $(2 - 2\bar{\lambda}, 1)$ slides left to $(0, 1)$, so the swept-out frontier traces out the two edges of the unit square that meet at the corner.

Q28. The blue curve (OK prediction). Moving from the lower-left to the upper-right along the blue curve, the agent increases TPR (catches more true positives) at the cost of also increasing FPR (more false alarms). This is the sensitivity–specificity trade-off. An agent with $0.5 < \bar{\lambda} < 1$ cannot achieve TPR = 1 and FPR = 0 simultaneously — the prediction technology is not accurate enough to perfectly separate the two states, so improving detection in one direction necessarily worsens it in the other.

Q29. Improving prediction. When AI "gets better," the ROC curve shifts toward the upper-left corner $(0, 1)$: from the red diagonal, toward the blue curve, and eventually toward the green L-shape. At every level of FPR, the agent can achieve a higher TPR. The area under the ROC curve (AUC) increases. In economic terms, the agent gets a better menu of error trade-offs, which increases the value of prediction for any decision context.